

DINUJA PERERA

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MACHINE LEARNING ENGINEER | NLP & PREDICTIVE MODELLING | DATA SCIENCE

Machine Learning Engineer with 2+ years of experience developing supervised, unsupervised, and deep learning models across predictive maintenance, NLP, and customer analytics. Recognized for building LSTM and ESN time-series models achieving low prediction error rates, with experience delivering cutting-edge research at WinNLP 2022, collocated with EMNLP's international conference. Combines strong Python development and ML pipeline expertise with a proven ability to contextualize complex data from diverse sources to deliver actionable intelligence for technical and non-technical stakeholders alike. **Tier 1 Global Talent Visa endorsed by UK Research and Innovation. No sponsorship required.**

- Python & ML Development
- NLP & Transformer Models
- Deep Learning (LSTM, ESN, RNN)
- SQL & Data Querying
- Scalable Data Pipelines
- Cross-Functional Collaboration

Technical Acumen: TensorFlow | Keras | PyTorch | PySpark | PyTest | Scikit-learn | XGBoost | Pandas
| NumPy | SQL | DuckDB | Azure Databricks | GitHub | Power BI | Tableau

PROFESSIONAL EXPERIENCE

Upskilling Residency | [Apziva](#) | United Kingdom

Jan. 2026 – Present

Completing an intensive upskilling program focused on CI pipelining, ML deployment capabilities, production-grade ML pipelines, and MLOps workflows including containerized model delivery.

- Built a multi-stage ML pipeline using Python predicting bank customer subscriptions to term deposits using demographic and campaign data. Optimized call timing and campaign strategy, saving 1,693 staff hours (212 working days) in unnecessary campaign calls, before segmenting remaining subscribers into 6 clusters using KMeans to determine optimal marketing strategies.
- Built a model to identify drivers of customer dissatisfaction in logistics surveys using ensemble methods on a highly imbalanced dataset. Achieved 0.77 recall for unhappy customers, identified 2 key questions driving dissatisfaction, and informed service improvement and survey redesign that expanded the questionnaire to focus on the most impactful factors.
- Developed an NLP-based candidate ranking system reducing manual review effort, benchmarking TF-IDF, Word2Vec, FastText, GloVe, and transformer models (BERT, E5-small) for candidate ranking. Evaluated results using cosine similarity, blended ranking metrics, and PCA/t-SNE visualizations. Integrated a large language model (Qwen2.5-0.5B-Instruct) from Hugging Face and tested prompt engineering techniques including few-shot, chat-format, and chain-of-thought prompting to improve output reliability

Machine Learning Engineer, Predictive Maintenance | [Uniper UK / Loughborough University](#)

Jun. 2024 – Present

Leads a Uniper–Loughborough University research partnership on gas turbine predictive maintenance, managing data from 1,000+ sensors per turbine. Developed foundational datasets by consulting engineers, reviewing manuals, and integrating cross-team sensor and fault data to support ML analysis. Research drives reduced National Grid penalties, improved compliance, and higher turbine uptime.

- Developed a digital twin using LSTM (TensorFlow) and ESN (built entirely from scratch without libraries) time-series models on approximately 200 turbine sensor streams identified as primary sources of false trip events, predicting outputs across multiple load conditions and achieving under 3% average prediction error. Applied SHAP, LOFO, and LIME for model explainability.
- Built ML models for sensor health monitoring using correlation-threshold methods (Spearman + Fisher Z transformation) to distinguish faulty sensor behaviour from normal operating variability, directly supporting efforts to reduce government penalties, environmental impact, and unplanned shutdown time.
- Developed a scalable missing-value strategy using null-percentage rules, sensor-family imputation, Random Forest regression, and flag columns to preserve full time-series integrity across fixed model inputs.
- Lead progress evaluation presentations to a range of technical and non-technical stakeholders, including government stakeholders, Uniper management, reliability engineers, plant operators, and academic supervisors. Collaborate with reliability engineers, plant operators and academic supervisors to translate operational requirements into ML solutions.
- Manage all datasets, models, and experiments in GitHub with structured documentation for reproducibility and knowledge transfer. Operate within Azure Databricks and joining data tables to support analysis across operational conditions.
- Apply K-Means clustering to identify distinct turbine rundown behaviour patterns supporting operational decision-making. Create different datasets for different load conditions to assess turbine performance under varying operational intensity.

Machine Learning Engineer | [CML Insight Inc.](#) | Texas, USA

Nov. 2021 – Jan. 2023

Developed causal machine learning solutions to identify key drivers of employee churn across large, sensitive employee datasets spanning multiple companies. Applied ML techniques to a dataset of 3,345 records and 206 features, focusing on meaningful feature behaviour to deliver actionable monthly insights to management.

- Engineered NLP features including tokenization, embeddings, and sentiment scores, along with novel features such as “thank you” frequency in incoming emails, enriching the feature space and improving churn prediction accuracy.
- Developed statistical methods using multimodal overlap scoring and distance-from-mean to measure class separation and identify features with consistent influence on churn outcomes.
- Maintained end-to-end ML pipelines with monthly model iteration cycles, incorporating new datasets, generating updated churn scores, and performing model drift and data drift checks to fine-tune model performance over time.
- Built reusable object-oriented Python pipelines for data preprocessing, feature engineering, and model evaluation. Used SQL queries in Python to combine datasets across projects. Operated within Linux-based systems.
- Collaborated with data scientists and engineers on end-to-end ML product development. Assessed feature importance and target leakage to improve model reliability. Onboarded and mentored interns on ML workflows.
- Analyzed and presented customer behaviour patterns for different customer groups to cross-functional teams, collaborating to review findings and inform business strategy.

Teaching Assistant | [University of Colombo](#) | Sri Lanka

Jun. 2019 – Nov. 2021

Delivered practical sessions and supervised projects for 100+ MSc and 200+ BSc students in Machine Learning & Neural Computing, Data Analytics, SQL, and Embedded Systems, reinforcing deep subject matter expertise across core data science and engineering disciplines. Taught dedicated practical sessions for SQL, demonstrating hands-on expertise in database querying and management.

Data Analyst Intern | [Brandix Apparel Limited](#) | Sri Lanka

Jun. 2019

EDUCATION**MSc Data Science – Distinction** | [University of Greenwich](#) | London, UK

2023

Key Project: Leveraging NLP for Enhanced Company Performance: Transformer-Based Sentiment Analysis

Implemented transformer-based sentiment analysis alongside topic modelling and aspect-based analysis to provide actionable insights for companies to improve products, services, and organizational aspects. Identified 8 positive and 2 negative entity-level sentiments (persons/departments) from Glassdoor reviews across 500 companies using aspect-based sentiment analysis. Topic modelling revealed that “good benefits & salary,” “work-life balance,” “supportive colleagues,” and “career growth” were the most discussed positive themes.

Coursework, MSc Data Science, Analytics & Engineering | [University of Moratuwa](#) | Sri Lanka

2021

Key Project: Benchmark NLP Algorithm for Hate Speech Detection

Developed 12 deep learning and transformer-based architectures and transformer-based approaches including RNNs, CNNs, BERT, RoBERTa, and hybrid architectures including CNN and LSTM. Leveraged project work to conduct further research and testing, demonstrating BART’s superior performance over RoBERTa models in binary (70%) and multi-class classification (58%) tasks. Research presented at the WinNLP Workshop co-located with EMNLP 2022.

Grad. Dip. Electronics, Telecom & Computer Engineering – Distinction | [Institution of Engineers](#) | Sri Lanka

2020

BSc in Physical Sciences (Computer Science, Physics, Applied Mathematics, and Statistics) – Second Class Honours

2019

[University of Colombo](#) | Sri Lanka**PROFESSIONAL DEVELOPMENT & PUBLICATIONS**

- Leadership & Management – CMI Level 5 Equivalent, Ashorne Hill Management College, UK.
- LinkedIn Learning Certifications: NLP, Advanced NLP, PySpark for Big Data, Azure DevOps Fundamentals, Generative AI Foundations, Introduction to LLMs, TensorFlow (CNN)
- Introduction to Artificial Intelligence with Python (CS50) – Harvard University (Expected Completion: April 2026)

Publication:

- Published “[Short Comparative Analysis on Pretrained BART and RoBERTa in Detecting Hate Speech on YouTube and Reddit Platforms](#)” at WinNLP Workshop co-located with EMNLP 2022, Abu Dhabi.